

Azamat Turganbayev

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Research Interests

Robot manipulation and tactile sensing, perception, and human-centered robotics, in particular rehabilitation exoskeletons and medical robotics.

Education

University of Minnesota (UMN), Twin Cities

Sep 2025 – Dec 2026 (expected)

MS in Robotics (GPA: 3.82/4.00)

Twin Cities, MN

- Coursework: Optimal Control and Reinforcement Learning, Nonlinear Optimization for ML and AI, Medical Robotics, Machine Learning Fundamentals, Robotics, Robot Vision, Introduction to Intelligent Robotics Systems, 3D Bioprinting

Nazarbayev University (NU, Leading University in Kazakhstan)

Aug 2018 – Jun 2023

BS in Robotics and Mechatronics

Astana, Kazakhstan

- Graduation Project: Shoulder Rehabilitation Robotic Exoskeleton
- Coursework: Electromechanical Systems, Mechanical Design with CAD, Image Processing, Linear Control Theory, Microcontrollers, Robotics: Kinematics and Dynamics, Industrial Automation

Publications

- [1] A. Niyetkaliyev, **A. Turganbayev**, M. Karasheva, R. Zhylkaidarov and Y. Turlybek, "Exploring the Potential of Four-Bar Linkages in Robotic Exoskeletons: A Comprehensive Review," in *Journal of Mechanical Design*, vol. 147, no. 10, April 2025, doi: [10.1115/1.4068107](https://doi.org/10.1115/1.4068107).
- [2] M. Karasheva, **A. Turganbayev**, A. Aimysheva and A. Niyetkaliyev, "Design of a 3D Printed Miniature Model for Human-Robot Mechanism Coupling for Shoulder Rehabilitation," *2023 8th International Conference on Robotics and Automation Engineering (ICRAE)*, Singapore, 2023, pp. 58-65, doi: [10.1109/ICRAE59816.2023.10458641](https://doi.org/10.1109/ICRAE59816.2023.10458641).
- [3] S. Omirbayev, I. Issa, Z. Kuangaliyev, **A. Turganbayev** and A. Niyetkaliyev, "The Use of Four-Bar Mechanisms in Robotic Exoskeletons," *2022 International Conference on Advanced Mechatronic Systems (ICAMechS)*, Toyama, Japan, 2022, pp. 149-156, doi: [10.1109/ICAMechS57222.2022.10003280](https://doi.org/10.1109/ICAMechS57222.2022.10003280).

Research Experience

MS Capstone Researcher (Python)

Fall 2026

ARDC Lab, UMN · Advisor: Ryan Caverly

Twin Cities, MN

- Capstone project: Center-of-Mass Self-Calibration for a Cable-Driven Spacecraft Motion Simulator (SPACESIM)
- Developing a method to identify the center of mass of SPACESIM's suspended 8-cable, 6-DOF chaser platform from the robot's own measurements, extending the lab's covariance-based self-calibration framework
- Set up the lab's Python CDPR simulator (SPARC-SIM), collected data in simulation, and started validating it ahead of tests on the small-scale prototype

Research Assistant (URScript, Python, ICP)

May 2026 – Present

Tactile Robotics Laboratory, NU · Supervisor: Zhanat Kappasov

Astana, Kazakhstan (now remote)

- Developed automated UR5 tactile exploration software and protocols for the lab's experiments; the collected data supports a paper in preparation for IROS 2027
- Sampled 3,000 surface points from the CAD model and aligned them to the robot base frame with an axis-constrained, translation-only ICP fit (~ 2.2 mm RMS)
- Wrote a custom Python inverse kinematics solver after the controller's built-in IK failed for the required poses
- Generated touch poses across 20 vertical strips with height-scaled tool tilt (7 to 14 degrees) for clearance
- Synchronized ATI Nano17 force data with the UR5's real-time pose stream via NI-DAQ, with automated press detection and peak-force logging
- Validated the pipeline in simulation (URSim in Docker) before running it on the physical robot

Senior Research Assistant *(SolidWorks, MATLAB/Simulink, QUARC)*

Nov 2023 – Apr 2025

Center of Excellence in Medical Robotics & Research, NU · Supervisor: Aibek Niyetkaliyev

Astana, Kazakhstan

- Led development of a full-scale 5-DOF shoulder rehabilitation exoskeleton from concept to validated hardware
- Combined a 2-DOF rotational four-bar linkage with a 3-DOF cable-driven parallel mechanism to keep the device aligned with the glenohumeral joint's center of rotation
- Tuned P/PI controllers for the linear and rotational drives in MATLAB/Simulink on QUARC real-time hardware
- Co-authored a review of four-bar linkages in robotic exoskeletons, published in the Journal of Mechanical Design (2025) [1]

Undergraduate Research Assistant *(SolidWorks, MATLAB, 3D Printing)*

Apr 2022 – Nov 2023

School of Engineering and Digital Sciences, NU · Supervisor: Aibek Niyetkaliyev

Astana, Kazakhstan

- Designed and built a 3D-printed miniature prototype of a robotic shoulder exoskeleton to validate the human-robot coupling mechanism
- Derived forward and inverse kinematics and validated them against the SolidWorks CAD model in MATLAB to within 1 to 2 mm
- Engineered a cable pretension mechanism that eliminated slack and derailment
- Published the results at ICRAE 2023 as second author [2]
- Co-authored a conference paper on four-bar mechanisms in robotic exoskeletons, presented at ICAMEchS 2022 [3]

Selected Projects

Automated Flashlight Assembly with UR5 *(RoboDK, Python)*

Spring 2026

College of Science & Engineering, UMN

Twin Cities, MN

- Programmed an autonomous three-part assembly (head, battery, endcap) via the RoboDK Python API, finishing in 1 min 39 s against a 110 s requirement
- Used joint-space motion for transits and Cartesian linear motion near contact across 15 defined targets
- Integrated pneumatic clamping with optical part sensing, and fastened parts by multi-stroke pre-threading followed by torque-controlled final tightening

TurtleBot3 Navigation & Obstacle Avoidance *(ROS 2, OpenCV, Python)*

Oct 2025 – Dec 2025

College of Science & Engineering, UMN

Twin Cities, MN

- Integrated and calibrated Raspberry Pi camera for ArUco marker-based tracking
- Built a ROS 2 pipeline to detect hazards and execute reactive maneuvers within 20 cm
- Fused camera-derived heading with LiDAR range in an ArUco follower node to hold target distance
- Validated in Gazebo and RViz before tuning LiDAR noise filtering on the physical robot

Honors & Awards

Bolashak International Scholarship

Sep 2025 – Dec 2026

Kazakhstan's nationally competitive merit scholarship, funding the MS in Robotics at the University of Minnesota

Full Scholarship, Nazarbayev University

Aug 2018 – Jun 2023

Covered the foundation year and the full BS in Robotics and Mechatronics

Skills

Programming Languages & Libraries: Python, C/C++, MATLAB, OpenCV

Robotics Frameworks & Tools: ROS / ROS 2, UR5, URScript, RoboDK, Gazebo, Linux, Git, Docker

Modeling & Simulation: SolidWorks, MATLAB/Simulink, QUARC, 3D Printing

Control Systems: PID Control Optimization, Real-Time Systems, Motion Planning